Introduction to Loop (Mesh) Analysis

Core Concept: If Nodal Analysis uses KCL for unknown **node voltages**, Loop Analysis uses KVL for unknown **loop currents**.

- Nodal analysis was developed by applying KCL at each non-reference node
- Loop analysis is developed by applying KVL around loops in the circuit
- Loop (mesh) analysis results in a system of linear equations which must be solved for unknown currents

Key Condition: Mesh Analysis applies **only to planar circuits**.

•Mesh analysis: applicable to planar circuit.

•A Mesh is a loop which does not contain any other loops within it

1.What is a Planar Circuit?

- A circuit that can be drawn on a flat plane without any wires crossing.

2.Defining a Mesh

- A **mesh** is a special kind of loop: it is a closed path that **does not contain any other loops inside it**.
- Think of it as the simplest possible loop—an "empty window pane" in the circuit.
- **Analogy:** A mesh is like a single field on a farm, not divided by any fences.

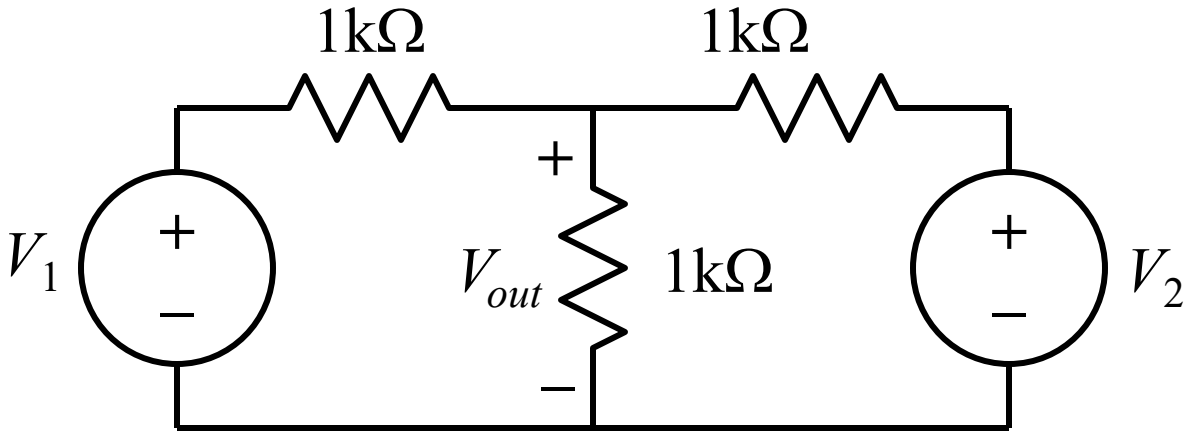
3.Mesh vs. Loop

- **All meshes are loops**, but not all loops are meshes.
- A loop can be large and contain smaller loops inside it. A mesh cannot.

•**Why it matters:** Using meshes guarantees we get the simplest set of equations to solve.

Another Summing Circuit

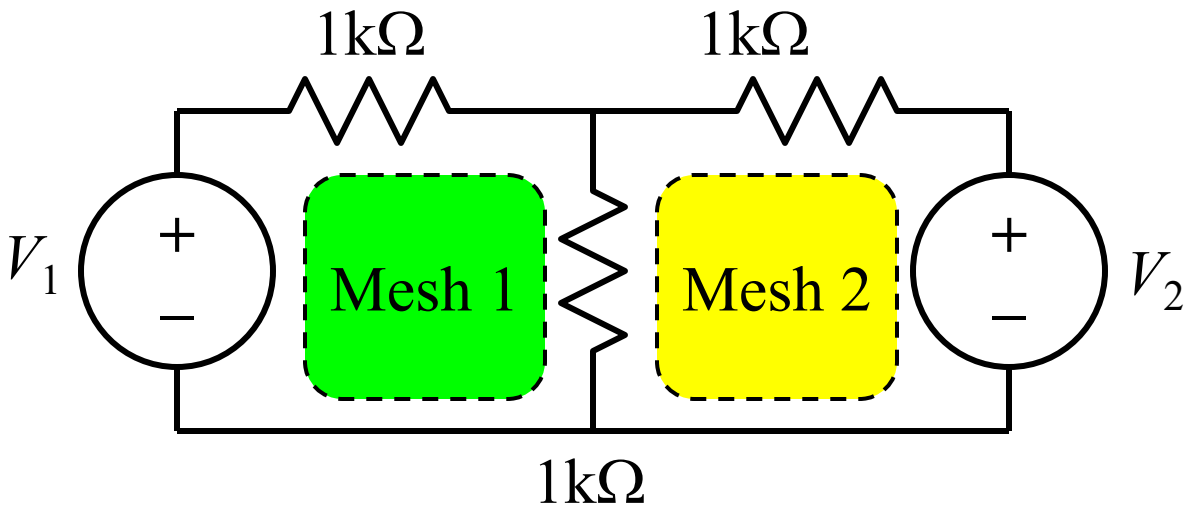
- The output voltage *V* of this circuit is proportional to the sum of the two input voltages V_1 and V_2



Steps of Mesh Analysis

1. **Identify mesh (loops).**
2. Assign a current to each mesh.
3. Apply KVL around each loop to get an equation in terms of the loop currents.
4. Solve the resulting system of linear equations for the mesh/loop currents.

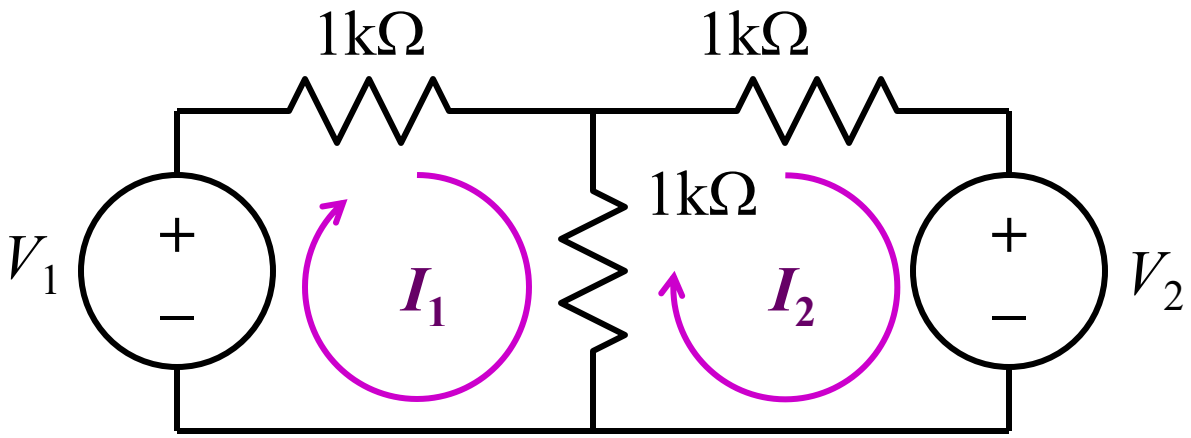
1. Identifying the Meshes



Steps of Mesh Analysis

1. Identify mesh (loops).
2. **Assign a current to each mesh.**
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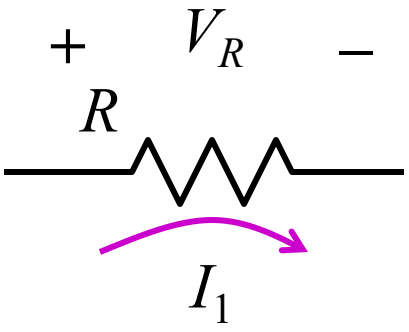
2. Assigning Mesh Currents

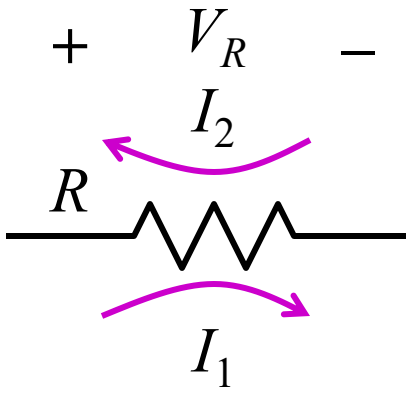


Steps of Mesh Analysis

1. Identify mesh (loops).
2. Assign a current to each mesh.
3. **Apply KVL around each loop to get an equation in terms of the loop currents.**
4. Solve the resulting system of linear equations for the mesh/loop currents.

Voltages from Mesh Currents

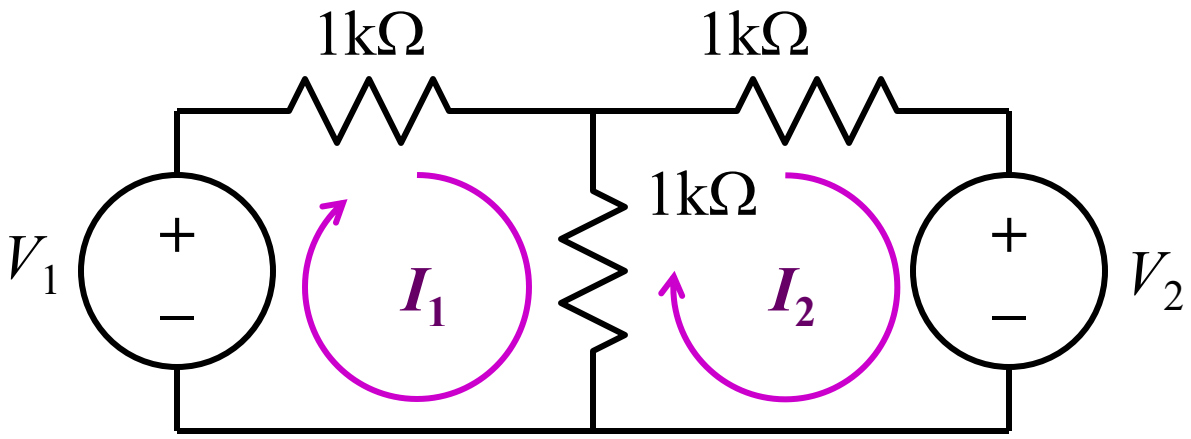




$$V_R = I_1 R$$

$$V_R = (I_1 - I_2) R$$

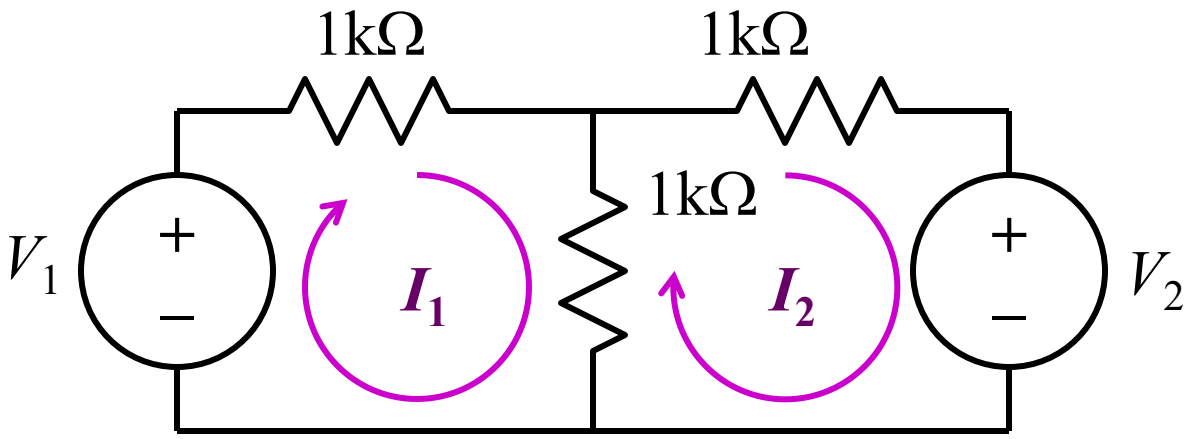
3. KVL Around Mesh 1



$$-V_1 + I_1\ 1\text{k}\Omega + (I_1 - I_2)\ 1\text{k}\Omega = 0$$

$$I_1\ 1\text{k}\Omega + (I_1 - I_2)\ 1\text{k}\Omega = V_1$$

3. KVL Around Mesh 2



$$(I_2 - I_1)\ 1\text{k}\Omega + I_2\ 1\text{k}\Omega + V_2 = 0$$

$$(I_2 - I_1)\ 1\text{k}\Omega + I_2\ 1\text{k}\Omega = -V_2$$

Steps of Mesh Analysis

1. Identify mesh (loops).
2. Assign a current to each mesh.
3. Apply KVL around each loop to get an equation in terms of the loop currents.
4. **Solve the resulting system of linear equations for the mesh/loop currents.**

Matrix Notation

- The two equations can be combined into a single matrix/vector equation

$$\begin{bmatrix} 1\text{k}\Omega + 1\text{k}\Omega & -1\text{k}\Omega \\ -1\text{k}\Omega & 1\text{k}\Omega + 1\text{k}\Omega \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}$$

4. Solving the Equations

Let: $V_1 = 7V$ and $V_2 = 4V$

Results:

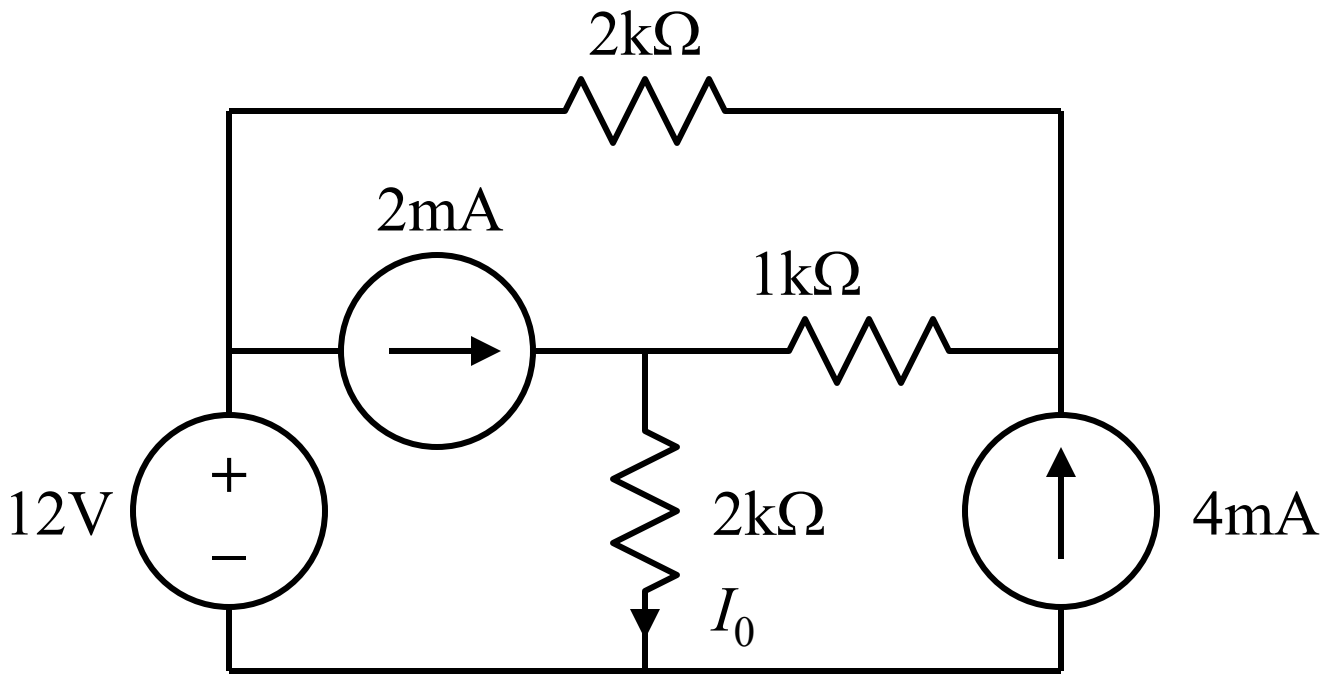
$$I_1 = 3.33 \text{ mA}$$

$$I_2 = -0.33 \text{ mA}$$

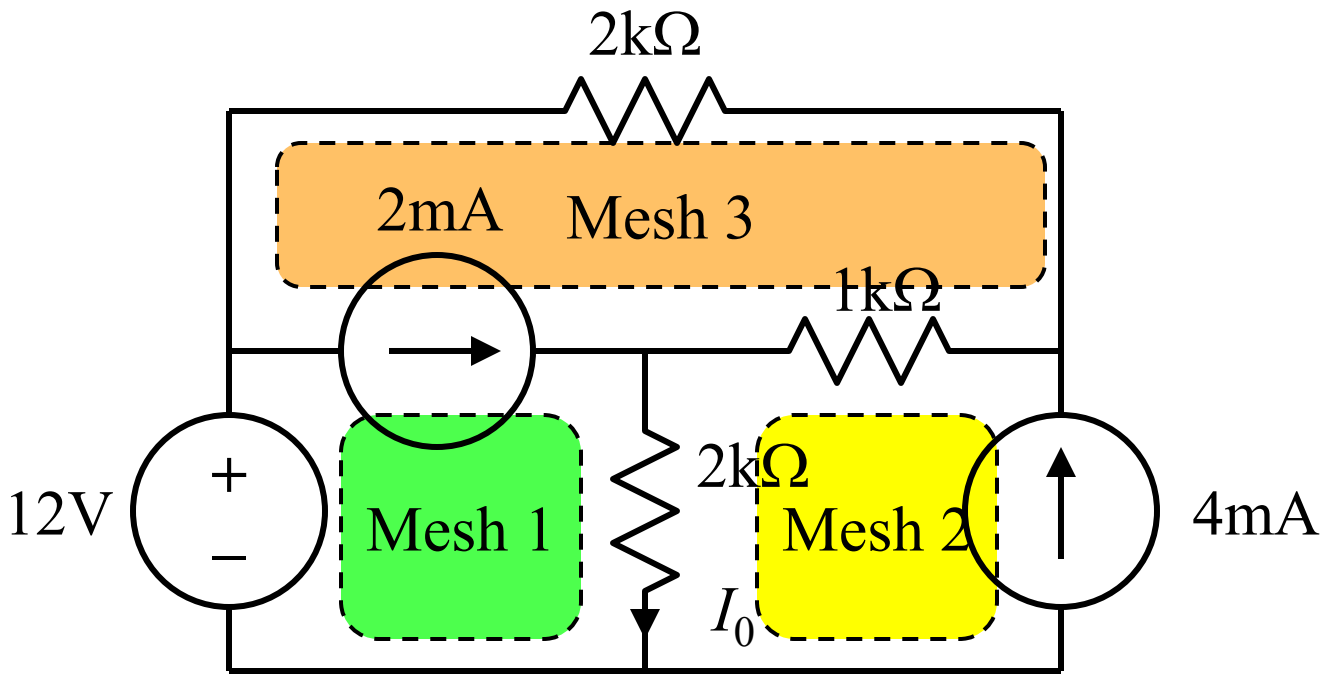
Finally

$$V_{out} = (I_1 - I_2)\ 1k\Omega = 3.66V$$

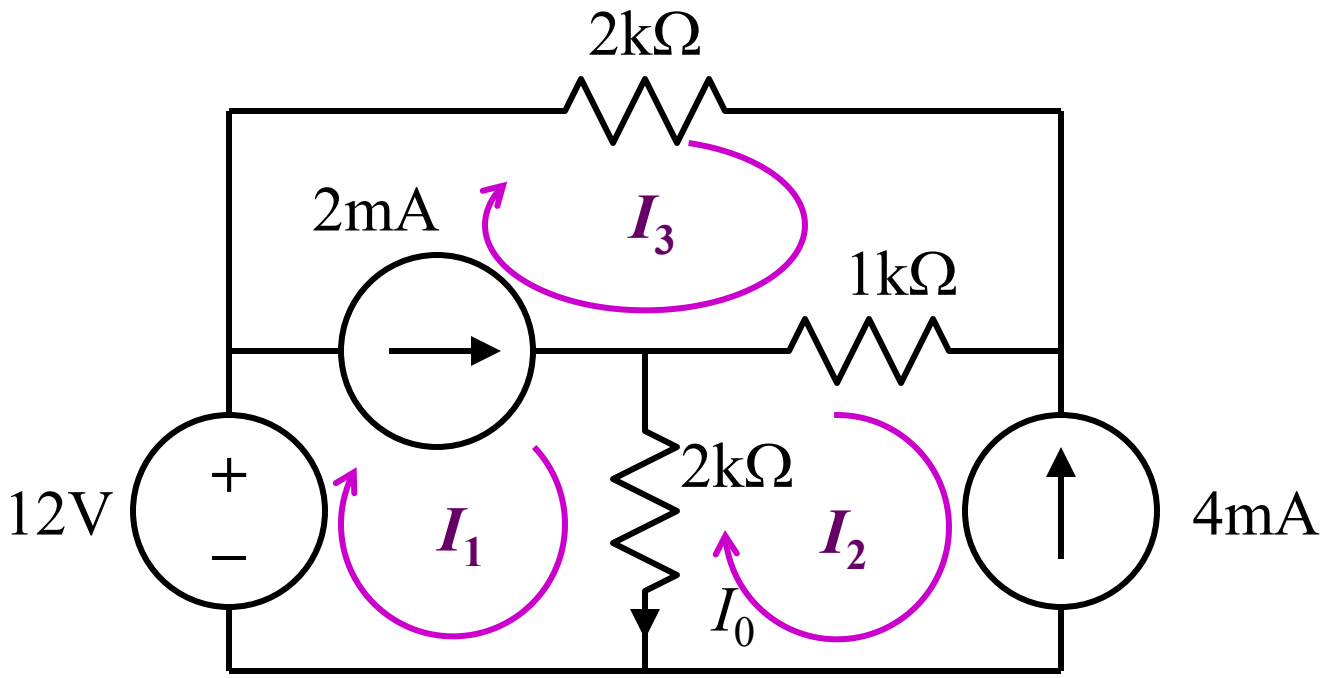
Another Example



1. Identify Meshes



2. Assign Mesh Currents



Current Sources

- The current sources in this circuit will have whatever voltage is necessary to make the current correct
- We can’t use KVL around any mesh because we don’t know the voltage for the current sources
- What to do?

Current Sources

- The 4mA current source sets I_2:

$$I_2 = -4 \text{ mA}$$

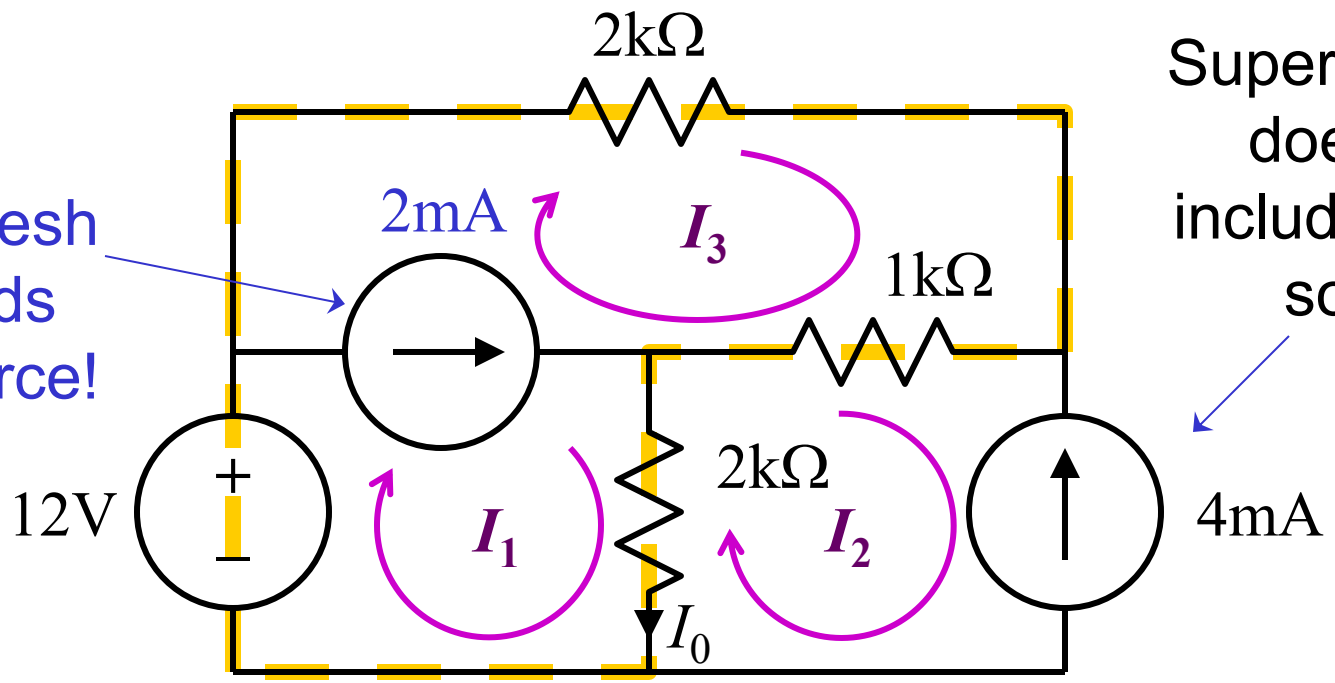
- The 2mA current source sets a *constraint* on I_1 and I_3:

$$I_1 - I_3 = 2 \text{ mA}$$

- We have two equations and three unknowns. Where is the third equation?

The supermesh!

The Supermesh surrounds this source!



The Supermesh does not include this source!

3. KVL Around the Supermesh

$$-12\text{V} + I_3\,2\text{k}\Omega + (I_3 - I_2)1\text{k}\Omega + (I_1 - I_2)2\text{k}\Omega = 0$$

$$I_3\,2\text{k}\Omega + (I_3 - I_2)1\text{k}\Omega + (I_1 - I_2)2\text{k}\Omega = 12\text{V}$$

Matrix Notation

- The three equations can be combined into a single matrix/vector equation

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 2\text{k}\Omega & -1\text{k}\Omega - 2\text{k}\Omega & 2\text{k}\Omega + 1\text{k}\Omega \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -4\text{mA} \\ 2\text{mA} \\ 12\text{V} \end{bmatrix}$$

Advantages of Nodal Analysis

- Solves directly for node voltages
- Current sources are easy
- Voltage sources are either very easy or somewhat difficult
- Works best for circuits with few nodes
- Works for any circuit

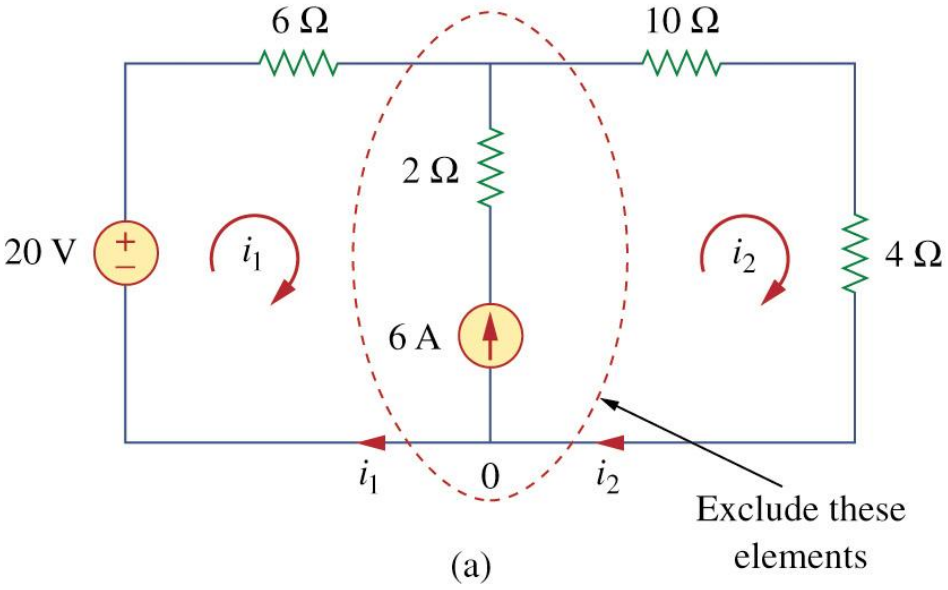
Advantages of Loop Analysis

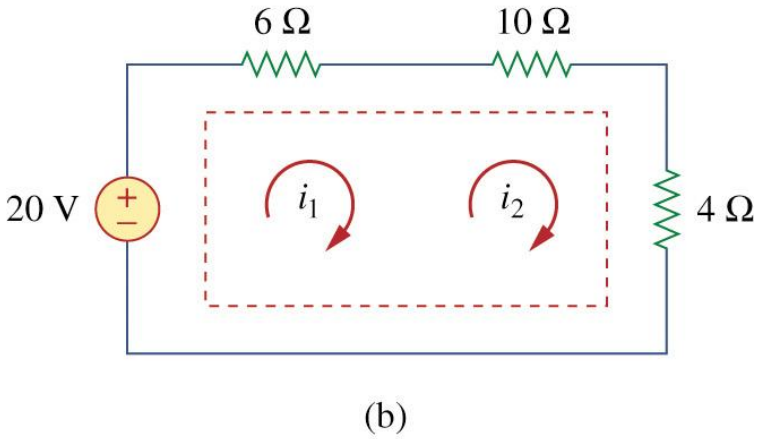
- Solves directly for some currents
- Voltage sources are easy
- Current sources are either very easy or somewhat difficult
- Works best for circuits with few loops

Disadvantages of Loop Analysis

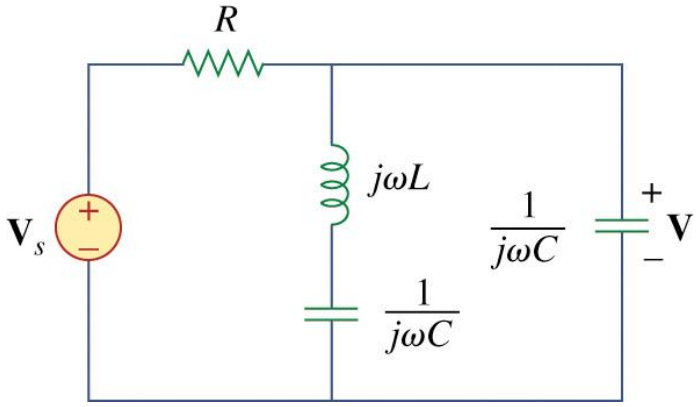
- Some currents must be computed from loop currents
- Does not work with non-planar circuits
- Choosing the supermesh may be difficult.

- a supermesh results when two meshes have a (dependent , independent) current source in common.





Using nodal analysis obtain **V** in the circuit shown.



NODE Equation

$$\frac{V - V_s}{R} + \frac{V}{j\omega L + \frac{1}{j\omega C}} + j\omega CV = 0$$

$$V + \frac{j\omega RCV}{-\omega^2 LC + 1} + j\omega RCV = V_s$$

$$\left(\frac{1 - \omega^2 LC + j\omega RC + j\omega RC - j\omega^3 RLC^2}{1 - \omega^2 LC} \right) V = V_s$$

$$V = \frac{(1 - \omega^2 LC)V_s}{1 - \omega^2 LC + j\omega RC(2 - \omega^2 LC)}$$